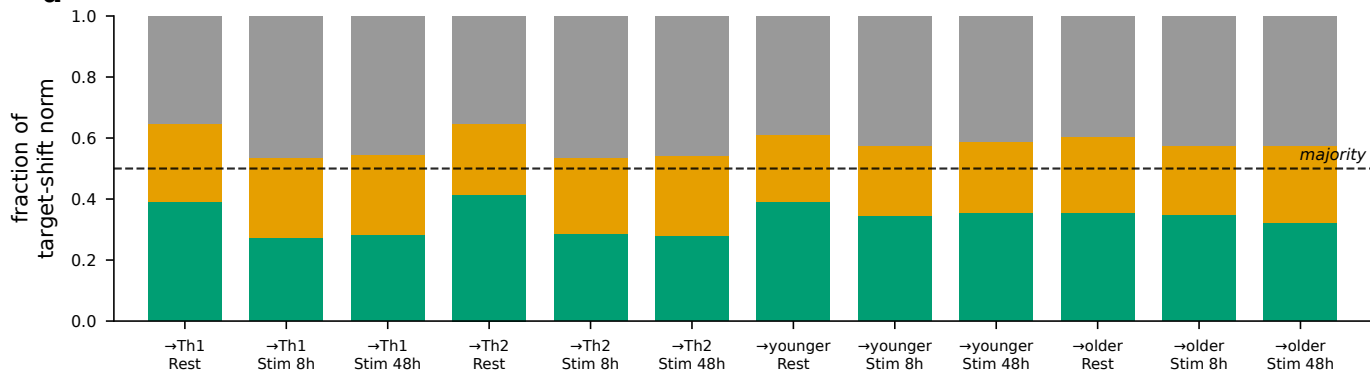


LOF (reachable by knockdown) GOF (activation-required) neither

Across all 12 atlas cells, knockdown is never the majority modality

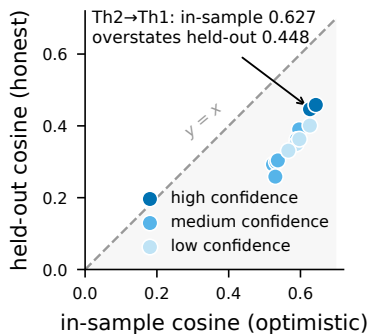
a

mean LOF fraction 0.34 · all held-out z 4.7 to 45.0



b

In-sample cosine overstates the honest value



c

Analytic null matches the shuffle null

